Uniform conditional moves and positivity thresholds in punctured-disk electrical networks

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Abstract

For every $n \geq 3$, we construct the four opposite-side conditional moves conjectured by Alexandr, Burks, Chepuri, and Commins for punctured-disk electrical networks. The moves are birational, preserve the response including the observed center, and are independent of the common exterior. We give a complete positivity test: a positive source admits the prescribed positive target exactly when n specified signed response minors are positive. Staircase separators with unique endpoint pairings identify these minors with positive multiples of center-source coordinates; at fixed circular-planar response, they describe a simplicial cone of center currents. For any positive baseline, vary the source edge conductance as ε or the source spike conductance as ε^{-1}. The positive realization set is exactly $(0, \varepsilon_*)$, so positivity cannot reenter. The selected minors have affine numerators, yielding an explicit threshold. At a finite threshold, the target converges to a network obtained by deleting center edges, with every other conductance finite and strictly positive.

1 Introduction

Alexandr, Burks, Chepuri, and Commins proposed four families of opposite-side conditional moves for electrical networks in a punctured disk [1, Section 5.2, Conjecture 4]. Depending on $n \bmod 4$, the move replaces an edge by a spike, a spike by a spike, a spike by an edge, or an edge by an edge. The center is an observed port. The problem is to realize the prescribed replacement with the same complete response and positive conductances.

We construct these moves for every $n \geq 3$ and determine their entire positive applicability along each natural degeneration ray. Fix positive conductances b on the common graph obtained by removing the selected feature. Insert a source edge of conductance ε, or a source spike of conductance ε^{-1}. The prescribed target is positively realizable precisely when

$$0 < \varepsilon < \varepsilon_*(b).$$

Thus positivity, once lost along the ray, never returns.

A uniform missing-connection minor determines the opposite feature; recovery on the common graph then gives a birational transformation. The map fixes the common exterior and restricts to a local move compatible with attachment along its complete interface (Theorem 2.1).

The second result is a complete positivity test in the original response coordinates (Theorem 2.2). A positive harmonic change of variables expresses the full response as a circular-planar response and a vector of center currents. At fixed circular-planar response, positive realizations form a simplicial cone. For each center edge, a staircase separator and a unique endpoint pairing produce a signed response minor that is a positive multiple of the corresponding center-source coordinate. These n minors describe all facets of the cone.

Finally, the source response varies by a rank-one update along the ray, so each selected minor has an affine numerator over a common positive denominator. Their simultaneous positivity yields the interval above and an exact threshold formula (Theorem 2.3). At every finite threshold, the target has a finite limit obtained by deleting center edges.

The classical inputs are circular-planar recovery and response cells [3], observed-center boundary recovery [1], and the all-minors matrix-tree identity [2]. Star insertion into a fixed critical circular-planar graph is already recoverable by [1, Theorem 6]; here we construct the prescribed opposite-side completion and prove independence from exterior parameters. Kenyon–Wilson [4] and Alman–Lian–Tran [5, Section 4] give positivity tests using selected circular minors. Our separating minors retain the internal observed center and describe the center-current fiber at a fixed circular-planar response, leading to the full positive range along each degeneration ray.

Section 3 constructs the rational move; Sections 4 and 5 establish its initial positive domain and locality. Sections 6 and 7 develop the center coordinates and positivity criterion. Section 8 proves non-reentry and the threshold formula, followed by an exact seven-sided example. Algorithmic details and supplementary calculations are collected in the appendices.

2 The four families and the main results

All indices on a polygon are taken in $\mathbb{Z}/n$. Write

$$L = \left\lfloor \frac{n+1}{4} \right\rfloor, \qquad n \in \{4L-1, 4L, 4L+1, 4L+2\}, \qquad L \geq 1.$$

Start with L concentric n-cycles, with vertices $w_{a,i}$, $0 \leq a < L$, numbered from the outside. Their edges are

$$w_{a,i}w_{a,i+1} \quad (0 \leq a < L), \qquad w_{a,i}w_{a+1,i} \quad (0 \leq a < L-1).$$

The outer observed vertices are $p_0, \ldots, p_{n-1}$. The outer modifications defining Π_n are as follows:

1. If $n = 4L-1$, set $w_{0,i} = p_i$ for every i.
2. If $n = 4L$, adjoin the spikes $p_i w_{0,i}$ for $i = 0, \ldots, 2L-1$; for the remaining indices set $w_{0,i} = p_i$.
3. If $n = 4L+1$, adjoin $p_i w_{0,i}$ for every i.
4. If $n = 4L+2$, adjoin all these spikes and the additional boundary edges

$$b_i = p_i p_{i+1}, \qquad -L \leq i \leq L.$$

In the last case the partial outer layer is a chain of $2L+1 = n/2$ edges on $2L+2$ boundary vertices, following [1, Figures 2 and 5]; for $n = 6$ it has three edges.

Adjoin an observed vertex z in the central face and all n edges $zw_{L-1,i}$. Denote this source graph by

$$X_n = \Xi_n, \qquad O = \{p_0, \ldots, p_{n-1}\}, \qquad B = O \cup \{z\}.$$

An inserted spike at p_t means that the old vertex p_t becomes an internal vertex, all its old edges remain incident to it, and a new observed vertex with the same label p_t is joined to it. Contracting a spike means that its internal endpoint becomes the observed vertex with the old boundary label.

The common graph H_n and the target X'_n are defined in Table 1. The added edges run along the outer boundary arc, and the new spikes are drawn outside the corresponding old polygon vertices. These changes preserve the punctured-disk embedding and the center port.

n	$X_n \longrightarrow H_n$	$H_n \longrightarrow X'_n$
$4L-1$	delete p_0p_1	insert a spike at p_{2L}
$4L$	contract $p_0w_{0,0}$	insert a spike at p_{2L}
$4L+1$	contract $p_0w_{0,0}$	add $p_{2L}p_{2L+1}$
$4L+2$	delete $b_0 = p_0p_1$	add $p_{2L+1}p_{2L+2}$

Table 1: The selected feature and its prescribed opposite-side replacement.

These choices use an endpoint spike of the partial layer when $n = 4L$ and the middle edge of that layer when $n = 4L+2$. Measure a boundary position by its cyclic index, assigning i to a spike at p_i and $i + \frac{1}{2}$ to the edge p_ip_{i+1}. In every row of Table 1, the target position minus the source position is $n/2$. This specifies the opposite side.

There are $n(2L-1)$ polygon and radial edges. Adding the boundary modifications gives $D = n(n-1)/2$ edges in Π_n. Consequently,

$$|E(X_n)| = |E(X'_n)| = d := \frac{n(n+1)}{2}, \qquad |E(H_n)| = d - 1. \tag{1}$$

For a weighted graph Y, let Λ_Y denote its response with positive diagonal Laplacian convention, defined in Section 3. Each edge retains its label through the specified deletion, contraction, or insertion.

2.1 Uniform construction and locality

Theorem 2.1 (Uniform local transformation)**.** *For every $n \geq 3$, the following statements hold.*

1. *There is a uniform finite rational algorithm for a birational map*

$$T_n : \mathbb{A}^{E(X_n)} \dashrightarrow \mathbb{A}^{E(X'_n)}$$

such that, with the same observed labels including z,

$$\Lambda_{X_n}(c) = \Lambda_{X'_n}(T_n(c))$$

as a rational identity. Its inverse is also given by rational recovery.

2. *There is an explicitly constructible nonempty Euclidean open box $U_n \subset \mathbb{R}_{>0}^{E(X_n)}$ on which every required denominator is nonzero and every target conductance is strictly positive. The formulas for the new boundary feature are positive for every strictly positive source input.*

3. *Remove common exposed boundary edges and common genuine boundary spikes by the deterministic procedure of Section 5. The remaining graphs G_n, G'_n have a common complete interface and a birational conditional move T_n^{loc} depending only on their own conductances. Its positive applicability region is nonempty and open. Common exterior conductances are unchanged. The move preserves response under any common attachment along the complete interface, whenever the resulting response is defined.*

The theorem gives the four opposite-side replacements of [1, Conjecture 4] for all n. Corollary B.2 in Appendix B describes the response geometry: near every positive response of H_n, the two images coincide with the same open side of $F_n = 0$.

2.2 A complete positivity criterion

Theorem 2.2 (Full-response positivity criterion)**.** *For every $n \geq 3$ and each target center edge $e_s = zw_{L-1,s}$, there are explicit disjoint ordered subsets $P_s, Q_s \subset B$, with $z \in Q_s$, and a sign $\sigma_s \in \{1, -1\}$, depending only on the graph, such that the signed minors*

$$f_s(M) = \sigma_s \det M_{P_s,Q_s} \tag{2}$$

give the following test. For every strictly positive source input c, the prescribed target has a strictly positive realization if and only if

$$f_s(\Lambda_{X_n}(c)) > 0 \qquad \text{for every } s. \tag{3}$$

The positive realization is unique.

The pairs and signs are explicit in Section 7.2. Theorem 7.3 identifies their minors with positive center-source coordinates over the common circular-planar cell. For the triangle, the formula in Appendix C shows how a center edge can reach zero while the new spike stays positive.

2.3 Positive applicability along degeneration rays

Fix $b \in \mathbb{R}_{>0}^{E(H_n)}$. The source vector $c_b(\varepsilon)$ keeps every inherited conductance equal to b and gives its selected edge conductance ε, or its selected spike conductance ε^{-1}. Define

$$M_b(\varepsilon) = \Lambda_{X_n}(c_b(\varepsilon)), \qquad A_n(b) = \{\varepsilon > 0 : M_b(\varepsilon) \in \Lambda_{X_n'}(\mathbb{R}_{>0}^{E(X_n')})\}. \tag{4}$$

Theorem 2.3 (Positive applicability and threshold)**.** *For every $n \geq 3$ and every $b \in \mathbb{R}_{>0}^{E(H_n)}$, there is $\varepsilon_*(b) \in (0, \infty]$ such that*

$$A_n(b) = (0, \varepsilon_*(b)).$$

With the minors of Theorem 2.2, one has

$$A_n(b) = \{\varepsilon > 0 : f_s(M_b(\varepsilon)) > 0 \text{ for all } s\}. \tag{5}$$

There are coefficients α_s, β_s determined by b such that

$$f_s(M_b(\varepsilon)) = \frac{\alpha_s + \beta_s \varepsilon}{d_b(\varepsilon)}, \qquad d_b(\varepsilon) = \begin{cases} 1, & \text{source edge}, \\ 1 + (M_0)_{00}\varepsilon, & \text{source spike}, \end{cases}$$

where $M_0 = \Lambda_{H_n}(b)$. In particular,

$$\varepsilon_*(b) = \min_{s:\, \beta_s < 0} \frac{\alpha_s}{-\beta_s}, \tag{6}$$

with the empty minimum equal to ∞. If $\varepsilon_*(b) < \infty$, the unique positive target conductances have finite limits there. Every noncenter edge has a strictly positive limit, and at least one center edge tends to zero.

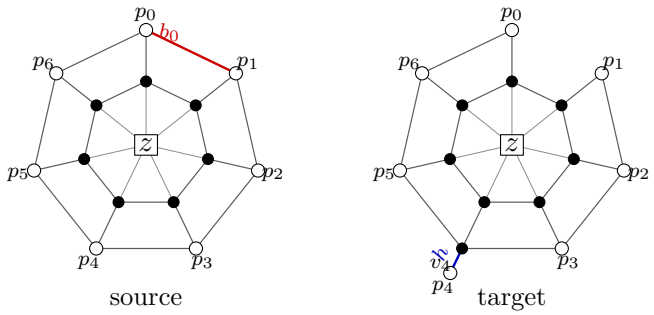


Figure 1: The full $n = 7$ source and target before common boundary stripping. The move deletes the outer edge $b_0 = p_0p_1$ and replaces the opposite outer vertex by an internal vertex v_4 with a new observed port p_4 attached by the spike h. White vertices, including the boxed centre port z, are observed; black dots are internal vertices.

3 Determinantal construction of the uniform move

A single circular minor F_n vanishes on the common graph H_n and is nonzero on either completion. Solving this vanishing condition determines the selected feature; recovery on H_n then determines all inherited weights. We first record the boundary identities and construct the missing connection.

Let Y be a finite loopless undirected graph with observed set B, internal set I, and conductances c_e. Its Laplacian is

$$K_{uv} = -\sum_{e:u\leftrightarrow v} c_e \quad (u \neq v), \qquad K_{uu} = \sum_{e\ni u} c_e.$$

Assume every connected component meets B. For strictly positive conductances the internal block is invertible, and

$$\Lambda_Y = K_{BB} - K_{BI}K_{II}^{-1}K_{IB}. \tag{7}$$

When I is empty, the determinant of the empty block is 1. Over $\mathbb{Q}$, formula (7) defines a rational map into

$$\mathcal{V}(B) = \{M = M^{\mathsf{T}} : M\mathbf{1} = 0\}, \qquad \dim \mathcal{V}(B) = \binom{|B|}{2}.$$

The entries M_{uv} with $u < v$, for a fixed ordering of B, are independent coordinates of this affine space. The vertex z belongs to B throughout.

We use the following standard circular-planar results. The graph Π_n is critical and has the standard medial matching [1, Corollary 1]. A critical graph remains critical after deletion of a boundary edge or contraction of a boundary spike [3, Lemmas 11.1–11.2]. Such a removal breaks some circular connection. Criticality is equivalent both to recoverability and to a lensless medial graph, and a critical medial matching determines the connection set [3, Section 7 and Lemmas 13.1–13.2]. Boundary conductance recovery with an observed center is [1, Lemma 10]. We record the formulas in our Laplacian convention.

3.1 Exact boundary formulas

For a boundary edge uv, put

$$E_{uv} = (e_u - e_v)(e_u - e_v)^{\mathsf{T}},$$

where e_u is the coordinate vector of the observed vertex. Adding conductance q adds qE_{uv} to the response.

Lemma 3.1. *If a spike of conductance h is inserted at observed vertex p of a network with response N, the response with the new observed vertex bearing the same label is*

$$S_{p,h}(N) = N - \frac{N_{:,p}N_{p,:}}{h + N_{pp}}. \tag{8}$$

The inverse operation is

$$N = M + \frac{M_{:,p}M_{p,:}}{h - M_{pp}}. \tag{9}$$

Both are rational identities on their indicated domains. For a positive spike, $h - M_{pp} > 0$.

Proof. The Schur pivot at the old observed vertex is $h + N_{pp}$, giving (8). Its p-column satisfies

$$M_{:,p} = \frac{h}{h + N_{pp}} N_{:,p}, \qquad h - M_{pp} = \frac{h^2}{h + N_{pp}} > 0.$$

These identities give (9). □

For any fixed square minor f, define the polynomials

$$\Delta^f_{uv}(M) = f(M + E_{uv}) - f(M), \qquad \beta^f_p(M) = f(M + M_{:,p}M_{p,:}) - f(M).$$

A determinant is affine in a rank-one update, so

$$f(M + sE_{uv}) = f(M) + s\Delta^f_{uv}(M), \tag{10}$$

$$f(M + sM_{:,p}M_{p,:}) = f(M) + s\beta^f_p(M). \tag{11}$$

Each response update used here preserves symmetry and zero row sums, since $E_{uv}\mathbf{1} = 0$ and $M_{p,:}\mathbf{1} = 0$.

3.2 Circular connections and groves

An ordered pair $P = (a_1, \ldots, a_k)$, $Q = (b_1, \ldots, b_k)$ of disjoint outer observed vertices is circular if $a_1, \ldots, a_k, b_k, \ldots, b_1$ occurs in circular order in one of the two orientations. A connection consists of vertex-disjoint paths from a_j to b_j, whose internal vertices are all internal vertices of the network.

Lemma 3.2. *Suppose deleting the central observed vertex z leaves a circular-planar graph, with the given cyclic order of the other observed vertices. For a circular pair P, Q avoiding z,*

$$(-1)^k \det M_{P,Q} = \frac{Z_{P|Q}}{\det K_{II}}, \tag{12}$$

where $Z_{P|Q}$ is the sum of weights of spanning forests whose boundary partition pairs a_j with b_j and leaves every other observed vertex in its own boundary class. For positive conductances this minor is nonzero, with sign $(-1)^k$, exactly when the connection exists.

Proof. The all-minors matrix-tree theorem [2] applies to

$$\det \begin{pmatrix} K_{II} & K_{IQ} \\ K_{PI} & K_{PQ} \end{pmatrix} = \det K_{II} \, \det M_{P,Q}.$$

Its omitted root sets are $B \setminus P$ and $B \setminus Q$. Each component contains one root from each set. An unused observed vertex belongs to both sets, so its component contains no other observed

vertex; this applies in particular to z. The pairing paths lie in the circular-planar graph with the unused observed vertices deleted. Their unique noncrossing pairing gives the sign $(-1)^k$ and (12). Conversely, a connection extends to a spanning forest with the required partition by contracting its paths and the unused observed vertices to roots and taking a rooted spanning forest. Every graph component contains such a root. For positive conductances every forest weight is positive, proving the stated equivalence. □

After boundary contractions, the same formula uses the current observed vertices.

3.3 The common missing connection

Define $F_n(M) = \det M_{P,Q}$ using Table 2. Integers in the table denote the labels p_i.

n	k	P	Q
$4L-1$	$2L-1$	$(-j)_{j=0}^{2L-2}$	$(1,\dots,2L-1)$
$4L$	$2L-1$	$(1,\dots,2L-1)$	$(-1,\dots,-(2L-1))$
$4L+1$	$2L$	$(1,\dots,2L)$	$(-1,\dots,-2L)$
$4L+2$	$2L+1$	$(0,-1,\dots,-2L)$	$(1,\dots,2L+1)$

Table 2: The common missing circular connection. All indices are modulo n.

Proposition 3.3. *The connection in Table 2 is absent in H_n and present in both X_n and X'_n. Therefore, for every positive input,*

$$F_n(\Lambda_{H_n}) = 0, \qquad (-1)^k F_n(\Lambda_{X_n}) > 0, \qquad (-1)^k F_n(\Lambda_{X'_n}) > 0. \tag{13}$$

Proof. Other observed vertices, including z, cannot be used as transit vertices. In H_n, the following sets of $k-1$ internal vertices separate the relevant sides:

1. For $n = 4L-1$ or $4L$, take columns $0, 2L$ on every inner layer $a = 1,\dots,L-1$. For $4L$, the corresponding outer vertices are unused observed vertices. For $4L-1$, the opposite outer vertex is unused and the edge p_0p_1 across the other cut is missing; p_0, which is an endpoint, stays on the P-side.

2. For $n = 4L+1$, take $w_{0,2L}$ and columns $0, 2L$ on the inner layers. The outer vertex in column 0 is the unused observed vertex p_0.

3. For $n = 4L+2$, take columns $0, 2L+1$ on all L layers. The remaining external boundary edges join vertices on the same side; the only source edge across the first cut has been deleted.

Their sizes are $2L-2, 2L-1, 2L$, respectively. After removing these vertices and the unused observed vertices, no P-to-Q connection can cross from one side to the other. Endpoints whose radial column meets the separator also require a separator vertex. Thus k disjoint paths cannot exist.

We now give the paths in the two completed graphs. For $4L-1$ and $4L+2$, number a pair by j, with endpoints p_{-j}, p_{j+1}. For $4L$ and $4L+1$, use endpoints p_j, p_{-j}. A path assigned to a polygon layer follows its two endpoint columns radially to that layer, traverses a polygon arc, and returns. Call the arc through column 0 the first arc, and the other arc the opposite arc.

Table 3 gives the layer assignments and the exceptional pair indices. Both s and a denote actual polygon layer numbers: the first three rows use only inner layers $1,\dots,L-1$, and the last row includes the outer polygon layer 0. The exceptional paths are:

- $4L-1$: source pair $j=0$ uses p_0p_1; target pair $j=2L-2$ goes through the old p_{2L}, now internal.
- $4L$: source pair $j=1$ goes through the old p_0; target pair $j=2L-1$ goes through the old p_{2L}.
- $4L+1$: source pairs $j=1,2L$ use the first and opposite arcs of the outer polygon. Target pair $j=2L$ uses the added external edge, and pair $j=2L-1$ uses the opposite outer polygon arc.
- $4L+2$: source pair $j=0$ uses b_0; target pair $j=2L$ uses the new external edge.

n	layer	source arc pair indices	target arc pair indices
$4L-1$	$s=1,\dots,L-1$	$(s, 2L-1-s)$ exception: 0	$(s-1, 2L-2-s)$ exception: $2L-2$
$4L$	$s=1,\dots,L-1$	$(s+1, 2L-s)$ exception: 1	$(s, 2L-1-s)$ exception: $2L-1$
$4L+1$	$s=1,\dots,L-1$	$(s+1, 2L-s)$ exceptions: $1, 2L$	$(s, 2L-1-s)$ exceptions: $2L-1, 2L$
$4L+2$	$a=0,\dots,L-1$	$(a+1, 2L-a)$ exception: 0	$(a, 2L-1-a)$ exception: $2L$

Table 3: At each layer the first index uses the arc through column 0, and the second uses the opposite arc. Exceptional pairs are assigned once per graph by the paths in the text.

For endpoints $(-j, j+1)$, the first arc has integer interval $[-j, j+1]$ through 0, and the opposite arc has interval $[j+1, n-j]$. For endpoints $(j, -j)$, these intervals are $[-j, j]$ and $[j, n-j]$. In each row of the table, the first index is strictly smaller than the second; the two intervals at that layer are consequently disjoint. As depth increases, the first index increases and the opposite index decreases. Their deeper endpoint columns lie outside both arcs already used on every shallower layer. The radial parts are therefore disjoint from those arcs and from each other.

The exceptional outer paths listed above obey the same separation. An external spike endpoint is distinct from the old polygon vertex; thus a direct external edge does not occupy a vertex of a polygon arc. Every pair is used exactly once, giving k vertex-disjoint paths. Indeed, the source pair indices are $0,\dots,2L-2$, $1,\dots,2L-1$, $1,\dots,2L$, and $0,\dots,2L$, respectively. For each target, the exceptional pair or pairs are the largest indices in the corresponding range, and the two layer sequences partition all the remaining indices. Empty inner-layer ranges for $L=1$ cause no exception. In particular, the first row of Table 2 then has $P=(0)$. Lemma 3.2 now gives (13). □

3.4 Recovery on the common graph

Recover the common graph from the outside inward, stopping when all innermost polygon vertices are observed.

Lemma 3.4. *The boundary recovery formulas of [3, Corollaries 4.3–4.4] and [1, Lemma 10] are as follows. Let a boundary edge or spike removal in the circular-planar graph obtained by deleting z break a circular connection with minor f. For the complete response M including z, its conductance is*

$$c_{uv} = \frac{f(M)}{\Delta^f_{uv}(M)} \qquad \text{for a boundary edge,} \tag{14}$$

$$c_p = M_{pp} - \frac{\beta^f_p(M)}{f(M)} \qquad \text{for a boundary spike at } p. \tag{15}$$

For a spike the broken connection avoids p. On positive network inputs all indicated denominators are nonzero.

Proof. A connection with endpoint p survives contraction of its spike, so a broken spike connection avoids p. By Lemma 3.2, the minor is nonzero before removal and zero afterwards. Substitute $M - cE_{uv}$ for an edge and (9) for a spike in the rank-one identities. This gives (14)–(15). For a positive spike $c - M_{pp} > 0$; the identities then give $\Delta^f_{uv} = f/c \neq 0$ or $\beta^f_p = -(c - M_{pp})f \neq 0$, respectively. □

Proposition 3.5. *There is a finite rational map*

$$\mathcal{R}_n : \mathcal{V}(B) \dashrightarrow \mathbb{A}^{E(H_n)}$$

with

$$\mathcal{R}_n(\Lambda_{H_n}(c)) = c$$

for every strictly positive conductivity on H_n. It is regular at every $\Lambda_{H_n}(b)$ with $b > 0$. The same procedure works if the center star is omitted altogether.

Proof. For $n = 4L + 2$, first recover and delete all remaining external boundary edges. In all four cases, next recover and contract all remaining extra spikes, so that every vertex of the outer polygon is observed. If that polygon is not the innermost one, recover and delete its remaining cycle edges. Every current outer observed vertex then has degree one: its only incident edge is the next inward radial edge. Recover and contract these radial spikes to expose the next layer. Repeat until all innermost polygon vertices are observed.

The center star is incident to the innermost layer, so the outer vertices being contracted retain degree one. At termination all vertices, including z, are observed. The remaining innermost cycle edges, possibly with one edge missing, and center edges are recovered as $-M_{uv}$; their endpoints uniquely determine the edge labels.

Deleting z initially gives a critical circular-planar graph: it is Π_n after one legal boundary deletion or contraction. Every subsequent step preserves criticality by the classical boundary removal results. Thus each selected feature has a broken circular connection. Enumerate circular pairs by size, ordered subset, and cyclic cut; take the first witness. This finite choice depends only on the graph, not on conductances. Apply Lemma 3.4 at each step, retaining original edge labels. This constructs $\mathcal{R}_n$ and recovers each conductance exactly.

All denominators are nonzero at every positive intermediate network. Thus the composed recovery map is regular at every $\Lambda_{H_n}(b)$, $b > 0$. Omitting the star changes only the last step: the residual cycle is read directly, with no center edges. □

The response image of H_n is Zariski-dense in the hypersurface $F_n = 0$; its algebraic description is given in Corollary B.1 of Appendix B.

3.5 Uniform reconstruction and the inverse transformation

Here and below, Δ, β without a superscript use the fixed minor F_n. For any specified completion of H_n, define its feature and stripped matrix by

$$\eta_{uv}(M) = \frac{F_n(M)}{\Delta_{uv}(M)}, \qquad N_{uv}(M) = M - \eta_{uv}(M)E_{uv}, \tag{16}$$

$$\eta_p(M) = M_{pp} - \frac{\beta_p(M)}{F_n(M)}, \qquad N_p(M) = M - \frac{F_n(M)}{\beta_p(M)} M_{:,p}M_{p,:}. \tag{17}$$

Both stripped matrices satisfy $F_n(N) = 0$. The target edge endpoints or spike label are those in Table 1. The source uses the edge p_0p_1 or the spike at p_0. The spike label is outside $P \cup Q$ in every required case.

3.5.1 Minor formulas for the opposite feature

The boundary determinant ratios of [3, Corollaries 4.3–4.4] take a particularly simple form for our fixed connection. Write P^-, Q^- for P, Q with their last entries removed, and set

$$F_n^-(M) = \det M_{P^-,Q^-}, \qquad C_{n,p}(M) = \det \begin{pmatrix} M_{P,Q} & M_{P,p} \\ M_{p,Q} & M_{pp} \end{pmatrix}.$$

The index p in $C_{n,p}$ is appended last in both lists.

Proposition 3.6. *For $n = 4L + 1$ or $4L + 2$, the new edge conductance is*

$$q_t = -\frac{F_n(M)}{F_n^-(M)}. \tag{18}$$

For $n = 4L - 1$ or $4L$, the new spike at $p = p_{2L}$ has conductance

$$h_t = \frac{C_{n,p}(M)}{F_n(M)}. \tag{19}$$

In both formulas $M = \Lambda_{X_n}(c)$. Their denominators are nonzero and their values are strictly positive for every $c > 0$.

Proof. The target edge joins the last matched pair in Table 2. Adding qE_t changes only the last diagonal entry of $M_{P,Q}$, by $-q$. Hence $\Delta_t(M) = -F_n^-(M)$, giving (18). For a spike, expansion of the bordered determinant gives the polynomial identity

$$C_{n,p}(M) = M_{pp}F_n(M) - \beta_p(M).$$

Equation (17) gives (19).

For $c > 0$, the full connection exists by Proposition 3.3. Dropping its last path gives the connection for P^-, Q^-. Lemma 3.2 therefore gives the signs $(-1)^k$ and $(-1)^{k-1}$ for F_n and F_n^-, respectively. Their negative ratio is strictly positive.

For the spike formula, temporarily count the outer vertex p as internal, retaining every other observed vertex, including z. Write $B' = B \setminus \{p\}$. The reduced response is

$$\widetilde{M} = M_{B',B'} - \frac{M_{B',p}M_{p,B'}}{M_{pp}}, \qquad M_{pp} > 0.$$

Connectedness of X_n gives $M_{pp} > 0$, and the original P-to-Q connection persists. Lemma 3.2 gives $(-1)^k \det \widetilde{M}_{P,Q} > 0$, and the block determinant yields

$$C_{n,p}(M) = M_{pp} \det \widetilde{M}_{P,Q}.$$

The numerator and denominator in (19) thus have the same nonzero sign. □

The new feature is therefore positive for every positive source input. The remaining positivity conditions will be expressed directly by the center-edge minors in Section 7.

3.5.2 The complete transformation

For $Y = X_n$ or X_n', assemble $\mathcal{R}_n(N(M))$ and $\eta(M)$ in the edge order of Y, and denote the resulting rational map by

$$\mathcal{J}_Y : \mathcal{V}(B) \dashrightarrow \mathbb{A}^{E(Y)}.$$

Proposition 3.7. *For every positive network on either completion Y,*

$$\mathcal{J}_Y(\Lambda_Y(c)) = c.$$

The maps Λ_Y and $\mathcal{J}_Y$ are birational inverses.

Proof. Removing the completion feature gives a positive actual network on H_n. Its minor is zero, whereas the completed minor is nonzero by Proposition 3.3. Equations (16)–(17), derived exactly as in Lemma 3.4, therefore recover the given feature and the stripped response. Proposition 3.5 then recovers all other conductances. Every denominator is nonzero on the positive input.

This identity on an open real parameter set is a rational identity. There are d parameters and d independent response coordinates. Every parameter is a rational function of those response coordinates, and each response coordinate is rational in the parameters. Thus

$$\mathbb{Q}(c_e : e \in E(Y)) = \mathbb{Q}(\text{independent entries of } \Lambda_Y).$$

The response coordinates are algebraically independent and generate the function field. Hence Λ_Y is birational onto $\mathcal{V}(B)$, with inverse $\mathcal{J}_Y$.

□

In particular the transformation in Theorem 2.1 is

$$T_n = \mathcal{J}_{X_n'} \circ \Lambda_{X_n}, \qquad T_n^{-1} = \mathcal{J}_{X_n} \circ \Lambda_{X_n'}. \tag{20}$$

Proposition 3.7 proves the complete response identity and the inverse identities. The formulas are uniform in n: the graph rules, the one minor, and the boundary-stripping order specify every target edge conductance.

4 Positive realization near the common degeneration

We now show that the rational transformation has a nonempty positive domain near every positive common baseline. This supplies the initial interval needed for the threshold theorem and the positive open set used in proving locality.

4.1 A transverse family at the common boundary

Fix any positive conductivity b on H_n, with response $M_0 = \Lambda_{H_n}(b)$. Use the degeneration ray $c_b(\varepsilon)$ of (4), and abbreviate its response as $M(\varepsilon) = M_b(\varepsilon)$. The source response is

$$M(\varepsilon) = M_0 + \varepsilon E_s \qquad \text{for an edge,} \tag{21}$$

$$M(\varepsilon) = M_0 - \frac{\varepsilon}{1 + \varepsilon (M_0)_{00}} (M_0)_{:,0} (M_0)_{0,:} \qquad \text{for a spike.} \tag{22}$$

Thus it is rational and regular at $\varepsilon = 0$.

Lemma 4.1. *Write $k = |P|$. The following numbers are nonzero:*

$$\gamma_{\mathrm{src}} = \begin{cases} \Delta_s(M_0), & \textit{source edge}, \\ -\beta_0(M_0), & \textit{source spike}. \end{cases}$$

They satisfy $(-1)^k \gamma_{\mathrm{src}} > 0$, and

$$F_n(M(\varepsilon)) = \begin{cases} \gamma_{\mathrm{src}}\varepsilon, & \textit{source edge}, \\ \gamma_{\mathrm{src}}\varepsilon/(1+\varepsilon(M_0)_{00}), & \textit{source spike}. \end{cases} \tag{23}$$

For a target edge $(-1)^k \Delta_t(M_0) > 0$. For a target spike $(-1)^{k+1}\beta_t(M_0) > 0$.

Proof. Since $F_n(M_0) = 0$, the rank-one identities give the displayed formulas. Adding any positive conductance on the source feature repairs the connection in Proposition 3.3. For a source edge its minor is $q\Delta_s(M_0)$. For a source spike of conductance h it is

$$-\frac{\beta_0(M_0)}{h + (M_0)_{00}}.$$

Lemma 3.2 gives the asserted nonzero signs. Apply exactly the same argument to the target feature, whose addition also repairs the connection. □

4.2 Exact conductances along the degeneration

The new feature has an exact rational formula along this family. For either fixed minor $A = F_n^-$ or $A = C_{n,p}$, put

$$D_s^A = \begin{cases} \Delta_s^A(M_0), & \text{source edge}, \\ (M_0)_{00} A(M_0) - \beta_0^A(M_0), & \text{source spike}. \end{cases}$$

The rank-one identities and (23) give

$$q_t(\varepsilon) = -\frac{\gamma_{\mathrm{src}}\varepsilon}{F_n^-(M_0) + \varepsilon D_s^{F_n^-}} \qquad (n = 4L+1,\ 4L+2), \tag{24}$$

$$h_t(\varepsilon) = \frac{C_{n,p}(M_0)}{\gamma_{\mathrm{src}}\varepsilon} + \frac{D_s^{C_{n,p}}}{\gamma_{\mathrm{src}}} \qquad (n = 4L-1,\ 4L). \tag{25}$$

For a source edge this follows by writing $A(M(\varepsilon)) = A(M_0) + \varepsilon\Delta_s^A(M_0)$. For a source spike the same calculation gives

$$A(M(\varepsilon)) = \frac{A(M_0) + \varepsilon D_s^A}{1 + \varepsilon(M_0)_{00}}.$$

All coefficients are rational functions of the chosen baseline b. Proposition 3.6 makes both feature formulas positive for every $\varepsilon > 0$.

Proposition 4.2. *For every $n \geq 3$ and every positive b on H_n, the target conductances $T_n(c_b(\varepsilon))$ are strictly positive, and all required denominators are nonzero, for sufficiently small $\varepsilon > 0$. Every inherited target conductance tends to its value in b. On a compact subset of the positive baseline space, this convergence and the choice of a sufficiently small positive ε can be made uniform.*

Proof. For a target edge, (16) and Lemma 4.1 give

$$\eta_t(M(\varepsilon)) = \frac{\gamma_{\mathrm{src}}}{\Delta_t(M_0)}\varepsilon + O(\varepsilon^2) > 0.$$

The stripped matrix tends to M_0. For a target spike they give

$$\eta_t(M(\varepsilon)) = \frac{-\beta_t(M_0)}{\gamma_{\mathrm{src}}}\varepsilon^{-1} + O(1) > 0.$$

Here the coefficient of $M_{:,t}M_{t,:}$ in the stripped matrix is $F_n(M(\varepsilon))/\beta_t(M(\varepsilon)) = O(\varepsilon)$, so the stripped matrix again tends to M_0.

The map $\mathcal{R}_n$ is regular at M_0 with value b. Its denominators therefore remain nonzero and all its output conductances tend to their strictly positive baseline values. The positive feature and these positive recovered conductances give a positive actual target graph. Its internal Laplacian block and the forward spike pivot are consequently nonzero. Both sides of the rational right-inverse identity $\Lambda_{X'_n} \circ \mathcal{J}_{X'_n} = \mathrm{id}$ are therefore defined at $M(\varepsilon)$, so this positive target has exactly that response. The source is positive for every finite $\varepsilon > 0$.

For uniformity, let b range over a compact positive set K. The normalized feature q_t/ε or εh_t, and the stripped matrix, are rational functions regular at every $(0, b)$, by the nonzero signs in Lemma 4.1. The same holds for the recovery expressions. The limiting scaled feature and inherited conductances have positive minima on K, as do the absolute values of the nonvanishing divisors in these regular expressions. Equation (23) controls the remaining factor ε in F_n. A finite cover of K by neighborhoods where these inequalities persist supplies one $\varepsilon_0 > 0$ for the whole set. □

In the notation of (4), Proposition 4.2 gives, for every $b > 0$, a $\delta(b) > 0$ such that

$$(0, \delta(b)) \subset A_n(b). \tag{26}$$

The local response geometry and an effective positive open box are established in Appendix B.

5 Locality and the common circular-planar cell

The common circular-planar cell has two roles: it allows the same exterior to be removed from both graphs, and it supplies the ordinary response coordinates for the positivity criterion. We establish the cell equality first, then recover the local move from its complete interface response.

5.1 The circular-planar target has the same medial matching

Remove z and its incident edges to obtain the circular-planar graphs C_n, C'_n and their common graph H_n°.

Lemma 5.1. *The graph C'_n is critical and has the same medial matching as $C_n = \Pi_n$.*

Proof. The star-free version of Proposition 3.5 recovers H_n°. The paths and the missing connection in Proposition 3.3 avoid z, so they are unchanged. For each positive network on C'_n, the same single-minor formulas recover its completion feature and its actual stripped response, and then recover every conductance of H_n°. Thus C'_n is recoverable for every positive conductivity. By [3, Lemma 13.2] it is critical.

It has $D = \binom{n}{2}$ edges. The critical medial criterion excludes closed strands and self-intersections and allows at most one intersection of each pair. Thus its $2n$ boundary endpoints belong to exactly

n strands. Each primal edge gives one intersection. Hence every pair of strands intersects once. For any strand, each of the other $n-1$ strands has one endpoint on each side of its endpoints. In the cyclic order of the $2n$ medial boundary points, those endpoints are consequently n positions apart. Its matching is $t_i \leftrightarrow t_{i+n}$. This is the standard matching of Π_n. □

By [3, Theorems 2 and 4], C_n and C'_n therefore have the same positive response cell, denoted $\mathcal{C}_n$. For either graph, the inverse response map on this cell is smooth [3, Theorem 4].

A boundary edge is *exposed* if its exterior side faces the intervening outer boundary arc in the star-free circular-planar graph. The common-removal procedure uses exposed edges with two observed endpoints and spikes of degree one at their observed endpoint in the full graph, retaining the punctured-disk embedding.

In a critical circular-planar graph an edge between consecutive observed vertices can be drawn exposed. Indeed, the region between it and the empty boundary arc has no other observed vertex. A connection using that region can only pair its two boundary endpoints, and can instead use the edge itself. Any other edge in the region would therefore be deletable without breaking a connection, contrary to criticality. The region contains no such edges, and the edge may be moved to its boundary arc.

Lemma 5.2. *Suppose two critical circular-planar graphs have the same labelled medial matching and a common exposed boundary edge at the same position, or a common boundary spike at the same observed label. After the same legal removal their labelled medial matchings and connection sets remain equal.*

Proof. Write t_{2i}, t_{2i+1} for the medial endpoints just before and after p_i. A spike at p_i has a crossing directly adjacent to these two endpoints. An exposed edge p_ip_{i+1} has a crossing directly adjacent to t_{2i+1}, t_{2i+2}. The common matching determines the other endpoints of the two strands. The same deletion or contraction resolves the same four endpoints in the same way; all other matched endpoints remain unchanged. Boundary removal preserves criticality [3, Lemmas 11.1–11.2]. The resulting matching determines the connection set by the critical medial classification [3, Section 7]. □

5.2 Common exterior conductances and the moving interface

Starting with the two graphs, repeatedly remove common exposed boundary edges and common genuine boundary spikes, checking legality in the full graphs containing z. Here common means that the inherited edge label and both current endpoint labels agree in the two graphs. In particular, a common spike has the same observed label and the same internal neighbor. At each iteration prefer edges, then spikes, and use the fixed edge-label order. For definiteness label extra boundary edges, cycle edges, radial edges, and extra spikes by their type, layer when present, and cyclic index, and fix the corresponding lexicographic order. Center edges are not candidates. The procedure stops when no common legal feature remains. It is finite and depends only on the two labelled embedded graphs. Let G_n, G'_n be the remaining graphs.

The circular-planar graphs at each stage remain critical and have the same matching, by Lemmas 5.1 and 5.2. Thus any circular connection broken on one side by the selected removal is also broken on the other. Lemma 3.4, applied to the common complete response including z, recovers the same conductance on both sides. Their updated complete responses also agree. This proves by induction, on the positive domain of T_n, that every removed exterior conductance is unchanged and that

$$\Lambda_{G_n}(c) = \Lambda_{G'_n}(d) \tag{27}$$

for the resulting core conductances. These identities are rational identities, since they hold on a nonempty real open set.

Record each removed edge in a common exterior graph E_n^{ext}. When a spike $p_i v$ is removed, the old physical vertex p_i remains in E_n^{ext}, and v becomes the new core interface vertex. The i-th matrix slot keeps its logical index, but its physical attachment is updated. Later radial removals update this interface in the same way. The center z remains observed throughout.

Reattaching E_n^{ext} along the complete current interface reconstructs the original source or target. For exterior conductances a, let $\mathcal{A}_a(N)$ be this rational operation on a core response N. It is obtained by adjoining the exterior Laplacian and eliminating the exterior internal vertices, or equivalently by reversing the recorded boundary operations. For positive core inputs it gives the complete original response.

5.3 The local transformation is independent of the exterior

Reattach the recorded exterior with all conductances equal to 1. Composing this operation with the complete reconstruction gives a left inverse for each core response map.

Proposition 5.3. *There are rational maps*

$$j_{G_n} = \mathrm{pr}_{G_n} \mathcal{J}_{X_n} \mathcal{A}_{\mathbf{1}}, \qquad j_{G'_n} = \mathrm{pr}_{G'_n} \mathcal{J}_{X'_n} \mathcal{A}_{\mathbf{1}}$$

that recover every positive core input from its complete interface response. The core component of $T_n(a, c)$ equals

$$T_n^{\mathrm{loc}}(c) = j_{G'_n}(\Lambda_{G_n}(c)), \tag{28}$$

as a rational identity, independently of a. This map is birational, with inverse $j_{G_n} \circ \Lambda_{G'_n}$.

Proof. Attach the fixed exterior E_n^{ext} with all conductances equal to 1 to any positive target core. The result is a positive network on X'_n. Its conductances are recovered by Proposition 3.7. Projecting to the core gives

$$j_{G'_n}(\Lambda_{G'_n}(d)) = d.$$

All divisions are defined on such a positive input. The same argument applies to the source core.

On the positive domain of the complete transformation, common exterior conductances are unchanged. Apply $j_{G'_n}$ to (27); the target core is positive, so the identity just proved gives (28). Its right side contains no exterior parameter. The positive domain is a nonempty Euclidean open set in all source parameters, so this equality is also a rational identity.

The full positive domain contains a family point with exterior parameters $\mathbf{1}$, by Proposition 4.2 with baseline $b = \mathbf{1}$. Choose a product neighborhood of this point inside the positive domain. Its slice at exterior parameters $\mathbf{1}$ is an open neighborhood of the core parameters. On this neighborhood the local formula is defined, the target core is positive, and (27) holds. Thus this response equality is a rational identity in the core parameters. Applying the source local left inverse gives

$$(j_{G_n} \Lambda_{G'_n}) \circ T_n^{\mathrm{loc}} = \mathrm{id}.$$

The two core graphs have the same number of edges, since each common step removes one edge on each side. A rational map between these equal-dimensional affine parameter spaces with a rational left inverse is birational, by the function-field argument of Proposition 3.7. The displayed map is its inverse. □

The selected source and target features remain in their respective cores. Formula (28) gives a direct algorithm: attach the recorded exterior with conductances 1 to the source core, compute its complete response, apply the target reconstruction, and retain only target core conductances. The rational sample constructed in Appendix B has exterior conductances $\mathbf{1}$. Its core restriction and Lemma B.3 give a computable positive open box for the local move.

Corollary 5.4. *Attach any common external network along the complete interface of G_n, G'_n. On the local positive applicability region, the response on all remaining observed vertices is the same whenever the response is defined. In particular an observed center port is preserved.*

Proof. Eliminate the local internal vertices first. The resulting matrices agree by (27). Adjoin the same exterior matrix and eliminate any remaining internal vertices. Associativity of Schur complements gives equality of the final complete responses. □

For $n = 3$ nothing is removed. For $n = 4$, remove the common spike at p_1. For $n = 5$, remove the common spikes at p_1, p_4. For $n = 6$, remove b_{-1}, b_1, then the common spikes at p_2, p_5. The remaining edge counts are respectively $6, 9, 13, 17$, giving the small local patches of [1] after cyclic relabelling. For $n = 7$, remove the four common outer edges $p_1p_2, p_2p_3, p_5p_6, p_6p_0$, then the radial spikes at p_2, p_6; the remaining patch has 22 edges. The common-removal procedure defines the local support for every n.

Together, (20), Propositions 3.6 and 4.2, Lemma B.3, Proposition 5.3, and Corollary 5.4 prove Theorem 2.1.

6 Harmonic coordinates retaining the center

The uniform move and its locality are now established. To determine its full positive domain, we separate the ordinary response from the currents at the center. We use the standard ground-state, or Doob, conductance scaling $c_{uv} \mapsto c_{uv}h_uh_v$; see [6, Proposition 2.1]. The boundary calculation below keeps the center currents explicit and yields the realization coordinates.

Let Y be either prescribed graph and let $C = Y - z$. Write $V = V(C)$ and set

$$O = B \setminus \{z\} = \{p_0, \dots, p_{n-1}\}, \qquad I = V \setminus O.$$

Let $S = \{w_{L-1,0}, \dots, w_{L-1,n-1}\}$, with actual vertex identifications understood. In particular, $S \cap O$ may be nonempty. For a symmetric zero-row-sum matrix M on B, put

$$A = M_{O,O}, \qquad j = A\mathbf{1} = -M_{O,z}, \qquad \mathcal{L} = A - \operatorname{diag}(j). \tag{29}$$

The pair $(\mathcal{L}, j)$ determines M: its center column is $-j$ and its center diagonal is $\mathbf{1}^t j$.

Lemma 6.1. *Suppose $M = \Lambda_Y(c)$ with $c > 0$. Let h be the harmonic potential with $h_O = 1$ and $h_z = 0$. Then $h_v > 0$ on V, and the conductances*

$$a_{uv} = c_{uv}h_uh_v \qquad (uv \in E(C))$$

have ordinary circular-planar response $\Lambda_C(a) = \mathcal{L}$.

Proof. The maximum principle and the connectedness of C give $h_v > 0$. Let K be the Laplacian with the row and column of z removed, and put $D_h = \operatorname{diag}(h)$. The off-diagonal entries of D_hKD_h are those of the Laplacian K_a of C. Its internal row sums vanish by harmonicity, and its outer row sums equal j. Consequently

$$D_hKD_h = K_a + \operatorname{diag}_O(j). \tag{30}$$

Taking the internal Schur complement, and using $h_O = 1$, gives $A = \Lambda_C(a) + \operatorname{diag}(j)$. □

For positive a on C, let $G = K_a[I, I]^{-1}$. When I is empty, expressions involving this block are omitted. Define the O-by-S exit matrix

$$\mathsf{H}_C(a)_{:,s} = \begin{cases} e_s, & s \in S \cap O, \\ (-K_a[O, I]G)_{:,s}, & s \in S \cap I. \end{cases} \tag{31}$$

Its entries are nonnegative and its columns sum to one. The column at an internal s is the distribution of the unit current injected there among the grounded outer ports.

Proposition 6.2. *For either prescribed graph, positive conductances are in bijection with pairs $(a, \mu) \in \mathbb{R}_{>0}^{E(C)} \times \mathbb{R}_{>0}^{S}$. The map from these coordinates is*

$$r_O = \mathbf{1}, \qquad r_I = \mathbf{1} + G\mu_I, \tag{32}$$

$$c_{uv} = a_{uv} r_u r_v, \qquad c_{zs} = r_s \mu_s. \tag{33}$$

Here μ is extended by zero off S. The corresponding full response is given by

$$A = \Lambda_C(a) + \operatorname{diag}(\mathsf{H}_C(a)\mu). \tag{34}$$

For $a > 0$ and $\mu \geq 0$, the same formulas give a network with strictly positive noncenter conductances and nonnegative center conductances, still satisfying (34). Its zero center conductances are exactly the zero coordinates of μ.

Proof. The inverse Dirichlet block G is entrywise nonnegative, so $r_I \geq 1$. All conductances in (33) are positive. Set $j = \mathsf{H}_C(a)\mu$. The internal equation in (32) and the definition of H_C give

$$K_a r = \mu - j^O,$$

where $j^O \in \mathbb{R}^V$ is the extension of j by zero from O to V. Comparison of diagonal and off-diagonal entries shows that the grounded Laplacian of (33) is

$$\operatorname{diag}(r) K_a \operatorname{diag}(r) + \operatorname{diag}_O(j).$$

Its Schur complement is (34). This calculation also holds for $\mu \geq 0$: then $r \geq 1$, and the grounded internal block is the positive definite matrix $\operatorname{diag}(r_I) K_a[I, I] \operatorname{diag}(r_I)$. Thus the response remains defined, including when every center edge is absent, and $r_s \mu_s = 0$ precisely when $\mu_s = 0$.

Conversely, use h from Lemma 6.1 and set $r = 1/h$, $a_{uv} = c_{uv} h_u h_v$, and $\mu_s = c_{zs} h_s$. Equation (30), compared on the diagonal, gives $K_a r = \mu - j^O$, hence (32). These operations are inverse to (33). □

Lemma 6.3. *For $C = C'_n$ and every $a > 0$, the matrix $\mathsf{H}_C(a)$ is invertible. More precisely, up to a fixed sign,*

$$\det K_a[V \setminus S, I] = \prod_{e \text{ radial or an outer spike}} a_e. \tag{35}$$

Proof. The all-minors matrix-tree theorem [2] expresses the determinant as a signed sum of forests having one root in S and one root in O per component. Their root paths give n disjoint paths between the outer ports and the innermost polygon. Each n-vertex layer is a separator, so these paths use every vertex of that layer, one per path. A horizontal step would use two vertices of a layer on one path and is impossible. The paths are therefore exactly the radial columns and their outer spikes. They cover all vertices, leaving no additional branches. An external edge joins

two outer roots and cannot occur. This is the unique contributing forest, giving (35). For $L = 1$, coincident outer and inner roots give zero-length paths; the same description applies, with empty products allowed.

If $\mathsf{H}_C(a)\mu = 0$, set $u_O = 0$ and $u_I = G\mu_I$. Then $K_a u = \mu$ on V, including its outer rows by (31). Thus $K_a[V \setminus S, I]u_I = 0$. Equation (35) implies $u = 0$ and then $\mu = 0$. □

Corollary 6.4. *Let M be symmetric with zero row sums, and suppose its matrix $\mathcal{L}$ from (29) lies in $\mathcal{C}_n$. Recover the unique positive a on C'_n with response $\mathcal{L}$, and put*

$$\mu = \mathsf{H}_{C'_n}(a)^{-1} j. \tag{36}$$

Then M has a positive realization on X'_n if and only if $\mu > 0$. Such a realization is unique.

Proof. Necessity follows from the inverse in Proposition 6.2 and uniqueness of the ordinary recovery. For sufficiency, reconstruct by (33). Its outer response block is $\mathcal{L} + \operatorname{diag}(j) = A$, and the zero row sums recover the same center column and diagonal. Lemma 6.3 and ordinary recovery also give uniqueness. □

7 A complete positivity test from response minors

The preceding coordinates reduce positivity to $\mu_s > 0$. We now read each of these signs from a minor of the original full response. The essential construction is a path family forced to use one specified center edge, with a fixed endpoint pairing.

7.1 Separating minors and center-source coordinates

For disjoint ordered subsets $P, Q \subset B$ of cardinality k, a full linkage means k vertex-disjoint paths joining P to Q, with no other observed vertex in a path interior. For the Laplacian sign convention used here, the all-minors forest identity is

$$\det M_{P,Q} = (-1)^k \frac{\sum_\pi \operatorname{sgn}(\pi) Z_\pi}{Z_{\mathrm{sep}}}. \tag{37}$$

The forest partition function Z_π pairs each P vertex with its assigned Q vertex and puts every other observed vertex in a separate component. The denominator separates all observed vertices. As in Lemma 3.2, this is the all-minors matrix-tree identity for the grounded Schur-complement minor, now with z allowed among the paired endpoints.

Definition 7.1. A separating pair for a center edge e_s is a pair (P_s, Q_s) as above, with $z \in Q_s$, such that the target has a full linkage, deleting e_s destroys all full linkages, and every full linkage has the same endpoint-permutation parity.

Lemma 7.2. *For a separating pair, let π_s have its common parity and set $\sigma_s = (-1)^k \operatorname{sgn}(\pi_s)$. Then f_s from (2) is strictly positive on every positive target response. It is zero on a target with e_s deleted, all noncenter conductances positive, and the other center conductances nonnegative.*

Proof. All terms in the numerator of (37) have the same sign. A full linkage extends to a forest of the indicated partition, so the numerator is nonzero for positive weights. After deletion there is no full linkage, hence no such forest. Further deletion of center edges cannot create a linkage. The Dirichlet block remains invertible because the positive graph C'_n is connected to the outer ports, even if z is isolated. □

Theorem 7.3 (Positive coordinates on the response fiber)**.** *For the separating pairs constructed in Lemma 7.4, over the cell $\mathcal{C}_n$ there are strictly positive functions $\kappa_s(\mathcal{L})$ such that*

$$f_s(M) = \kappa_s(\mathcal{L})\mu_s, \qquad \mu = \mathsf{H}_{C'_n}(a)^{-1} j. \tag{38}$$

Consequently a symmetric zero-row-sum M with $\mathcal{L} \in \mathcal{C}_n$ has a positive target realization exactly when $f_s(M) > 0$ for every s.

Proof. Fix $\mathcal{L}$, and hence the recovered a and its invertible exit matrix. Since P_s and Q_s are disjoint, the outer-to-outer entries of the minor are off-diagonal entries of $\mathcal{L}$. Its only column involving j is the center column $-j$. Thus f_s is homogeneous linear in j, and hence in μ.

Take $\mu_s = 0$ and all other μ coordinates positive. Formulas (32)–(33) still construct a network with all noncenter conductances positive and exactly the center edge e_s deleted. Lemma 7.2 gives $f_s = 0$. A linear function vanishing on this relatively open part of the coordinate hyperplane must be a scalar multiple of μ_s. Taking all μ coordinates positive shows that this scalar is strictly positive. Corollary 6.4 now proves the claimed equivalence. □

Thus the required positive response region, over a fixed $\mathcal{L}$, is the simplicial cone $\mathsf{H}_{C'_n}(a)\mathbb{R}^S_{>0}$ in the variable j. The minors are positively rescaled coordinates of this cone. We next construct the separating pairs for every prescribed graph.

7.2 Uniform staircase separators

7.2.1 Endpoint sets

Let D be the target's set of outer spike indices and let E index its external edges $p_i p_{i+1}$. Directly from Table 1, these data are as in Table 4. The vertices $w_{j,i}$ retain their labels after any boundary identification.

n	D	Other target boundary data
$4L-1$	$\{2L\}$	missing polygon edge $w_{0,0}w_{0,1}$
$4L$	$\{1,\dots,2L\}$	none
$4L+1$	$(\mathbb{Z}/n)\setminus\{0\}$	$E = \{2L\}$
$4L+2$	$\mathbb{Z}/n$	$E = \{-L,\dots,-1\}\cup\{1,\dots,L\}\cup\{2L+1\}$

Table 4: Boundary data for the prescribed targets.

Fix s and write

$$a = s - L, \qquad b = s + L, \qquad \delta_a = \mathbf{1}_{a\in D}, \quad \delta_b = \mathbf{1}_{b\in D}.$$

In this section a, b are indices, not conductance vectors. Intervals below follow increasing cyclic order, with the length specified by the displayed integer endpoints. Choose the consecutive outer block

$$P_s = \{s-L+1,\dots,s+L-1\}\cup(\{a,b\}\cap D), \qquad k = 2L-1+\delta_a+\delta_b. \tag{39}$$

Initially take the outer part of Q_s to be all remaining indices, except a and b when they have no spike. Make the corrections in Table 5, and then include z in Q_s. Finally order both sets by $p_0,\dots,p_{n-1},z$ for computing the determinant and its permutation sign.

The sets are disjoint and have equal cardinality k. For $n = 4L$, exactly one of a, b lies in the consecutive half-circle D. For $n = 4L+2$, the sets E and $E+(2L+1)$ are complementary, so exactly

n	Condition	Correction to the outer part of Q_s
$4L-1$	$a=2L$ or $b=2L$	Add back the other, nonspiked endpoint among a, b.
$4L$	all s	No correction.
$4L+1$	$a=0$ $b=0$	Delete $2L+1$. Delete $2L$.
$4L+2$	$a-1\in E$ $b\in E$	Delete $a-1$. Delete $b+1$.

Table 5: The boundary corrections in the staircase construction. Unlisted cases require no correction.

one of the last two corrections applies. For $n=4L-1$, the exceptional centers are $s=L, 3L$; for $n=4L+1$, they are $s=L, 3L+1$. Counting the added or omitted endpoints verifies the equality in each case. The outer part of Q_s is a consecutive block separated from P_s.

7.2.2 The separator and the path family

Define

$$T_s=\{w_{j,s-(L-j)}, w_{j,s+(L-j)} : 1\le j<L\} \cup \{w_{0,a} : a\in D\}\cup\{w_{0,b} : b\in D\}. \tag{40}$$

The displayed vertices are distinct and $|T_s|=k-1$. Between the two staircases, row j has the indices

$$s-(L-j)+1,\dots,s+(L-j)-1. \tag{41}$$

The last row contains only $w_{L-1,s}$.

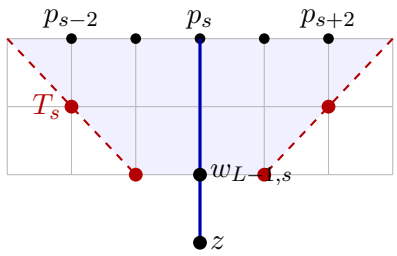


Figure 2: The staircase geometry for $L=3$. The full polygon grid is unwrapped into rows to locate the separator, and only the central center edge is drawn. The actual boundary edges, spikes, and port identifications are specified in Tables 4 and 5.

Lemma 7.4. *For every $n\ge 3$ and every center index s, the sets (39) and Table 5 form a separating pair for e_s. In fact the endpoint pairing is unique, and pairs p_s with z.*

Proof. We prove separation, existence, and uniqueness in turn.

Separation. Remove the unselected observed vertices, which cannot occur on a linkage. Each horizontal or radial edge leaving (41) meets T_s, except possibly at the two outer endpoints a, b. The boundary rules handle these exceptions as follows. For $4L-1$, ordinarily both endpoints are unselected. At an exceptional center one is a spike endpoint in T_s; the other is in Q_s, but its only polygon edge into the wedge is the missing edge 0–1. When $L\ge 2$, its inward radial edge also lies

outside the wedge; when $L = 1$, there is no interlayer radial edge. For $4L$, the nonspiked endpoint is unselected and the spiked one is blocked by T_s.

For $4L+1$, the added external edge can cross the wedge boundary only when $a = 0$ or $b = 0$. The corresponding correction removes its exterior observed endpoint. For $4L + 2$, the complementary half-circle property of E leaves exactly one external edge crossing the two sides; its exterior observed endpoint is again removed. Other external edges join observed points on the same side and do not bypass the separator.

The only innermost vertex inside the wedge is $w_{L-1,s}$. Deleting T_s and e_s therefore separates P_s from Q_s. Since $|T_s| = k - 1$, there is no full linkage after deleting e_s.

Existence. Join p_s to z by its radial column, including its outer spike if present and then e_s. Put $r_+ = L - 1 + \delta_b$ and $r_- = L - 1 + \delta_a$. List $Q_s \setminus \{z\}$ as $q_1, \ldots, q_{k-1}$ in increasing cyclic order, starting immediately after the rightmost endpoint p_{s+r_+} of the P_s block and skipping unselected ports. The explicit pairing is

$$p_s \longleftrightarrow z, \qquad p_{s+r} \longleftrightarrow q_{r_+-r+1}\ (1 \le r \le r_+), \qquad p_{s-r} \longleftrightarrow q_{r_++r}\ (1 \le r \le r_-). \tag{42}$$

Since $r_+ + r_- = k - 1$, these assignments use all endpoints once. On each side the outermost P_s endpoint is paired with the nearest Q_s endpoint.

For $p_{s\pm r}$, descend along its column to layer $L - r$, follow that polygon in increasing cyclic index order for p_{s+r} and decreasing cyclic index order for p_{s-r} until reaching its assigned Q_s column, and ascend to the observed endpoint. Include outer spikes at the ends where necessary. On each side, nested endpoint intervals have strictly increasing depths. A deeper horizontal segment lies below the radial ends of a shallower path, while its own radial ends lie outside the shallower horizontal interval. Thus the paths are disjoint on that side. The left and right intervals are disjoint and both avoid the central column.

A depth-zero path occurs only at a spiked endpoint a or b. Normally it uses one polygon edge. When an adjacent Q_s endpoint was omitted in the $4L + 1$ or $4L + 2$ case, it uses two; the intermediate vertex is the internal end of an unused spike, not an observed vertex. In the $4L - 1$ exceptional cases, the depth-zero path starts at $2L$ and does not use the missing 0–1 edge. All other horizontal segments are internal. This constructs a full linkage.

Uniqueness of pairing. Add $w_{L-1,s}$ to T_s, obtaining a separator of size k. Any full linkage must use every separator vertex and exactly one per path.

First let $L = 1$, so $n \in \{3, 4, 5, 6\}$, and put $v = w_{0,s}$. The two neighboring polygon indices are $a = s - 1$ and $b = s + 1$. Each spiked neighbor belongs to T_s. For $n = 4, 5$, each nonspiked neighbor is an unselected observed vertex; for $n = 6$ both neighbors are spiked. For $n = 3$, a nonspiked neighbor is either unselected or is the exceptional endpoint added back to Q_s; in the latter case its edge to v is the missing 0–1 edge. Thus every polygon exit from v is blocked by T_s, an unselected observed vertex, or a missing edge. If $s \in D$, the only remaining directions at v are its spike to p_s and its center edge to z. Since a linkage path cannot use a second separator vertex, the path through v is p_s, v, z. If $s \notin D$, then $v = p_s$, and the same restriction forces its path to be the edge $p_s z$; other observed vertices cannot be path interiors, and no external edge is incident to such an unspiked p_s in these four targets. The remaining outer endpoints have the unique reverse-order pairing by planar disjointness. This proves the claim for $L = 1$.

Now let $L \ge 2$. A path through $w_{L-1,s}$ must end at z: its radial neighbor is inside the wedge and both innermost polygon neighbors belong to T_s. Reaching any other outside endpoint would require a second separator vertex. All separator vertices are internal. Work in the embedded graph after deleting the unselected observed vertices and their incident edges. Consecutive vertices on either staircase are opposite corners of an interlayer quadrilateral. Join them by an arc whose

interior lies in that open face. At the bottom, join $w_{L-1,s-1}$ to $w_{L-1,s}$ and then to $w_{L-1,s+1}$ by arcs in the two adjacent faces toward layer $L-2$, with their interiors off the innermost polygon edges. The left and right arcs have disjoint interiors: their face interiors are disjoint, and the two bottom arcs meet only at $w_{L-1,s}$.

Extend the two outer ends to the disk boundary on the respective sides of the P_s block. The missing edge, deleted unselected endpoints, and external-edge corrections checked in the separation step provide the required gaps in the remaining graph. Choose the extensions through these gaps, keeping any corner spikes with endpoints in P_s on the P_s side. The resulting simple arc meets the remaining graph exactly at $T_s \cup \{w_{L-1,s}\}$, in the order down the left staircase, through the bottom vertex, and up the right staircase. Together with the outer boundary arc containing P_s, it bounds a disk.

Truncate each linkage path at its first separator vertex. These disjoint segments lie in that disk and end at all k separator vertices. Two segments with alternating endpoints on the disk boundary would intersect, so the matching to the separator is forced by boundary order. Its exits are the left staircase, the bottom vertex, and the right staircase. There are $L-1+\delta_a$ exits to the left of the bottom vertex and $L-1+\delta_b$ to its right, exactly the numbers of P_s endpoints on the two sides of p_s. Hence the path ending at z starts at p_s in this case as well. The remaining endpoints lie in two separated outer arcs, for which planar disjointness forces the unique reverse-order pairing. This proves uniqueness and therefore the required common permutation parity; the pairing is (42). □

Proof of Theorem 2.2. For a positive source, Lemma 6.1 places $\mathcal{L}$ in the positive response cell of C_n, which equals $\mathcal{C}_n$ by Lemma 5.1 and the circular-planar response-cell theorem. Lemma 7.4 constructs the separating pairs. Theorem 7.3 gives the stated criterion, and Corollary 6.4 gives uniqueness. □

8 Non-reentry and the positivity threshold

The full-response criterion turns the ray problem into simultaneous sign conditions. The rank-one response formulas make these conditions affine after multiplication by a common positive denominator.

Lemma 8.1. *Let $M_0 = \Lambda_{H_n}(b)$. For the two source feature types,*

$$M_b(\varepsilon) = \begin{cases} M_0 + \varepsilon v v^t, & v = e_0 - e_1, \quad \textit{source edge}, \\ M_0 - \dfrac{\varepsilon}{1+m\varepsilon} u u^t, & u = (M_0)_{:,0},\ m = (M_0)_{00}, \quad \textit{source spike}. \end{cases} \tag{43}$$

Here $m > 0$. Every fixed response minor is affine in ε in the edge case, and has an affine numerator over $1+m\varepsilon$ in the spike case.

Proof. The response formulas are (21) and (22); connectedness gives $m > 0$. Column multilinearity makes every minor affine in the rank-one update parameter. Substituting $-\varepsilon/(1+m\varepsilon)$ gives the asserted affine numerator in the spike case. □

Proof of Theorem 2.3. The source is positive for every $\varepsilon > 0$, so Theorem 2.2 gives (5). Lemma 8.1 gives its affine numerators and common positive denominator.

By (26), each numerator $\alpha_s + \beta_s\varepsilon$ is positive for sufficiently small positive ε. Thus $\alpha_s \geq 0$, and $\alpha_s = 0$ requires $\beta_s > 0$. If $\beta_s < 0$, its positivity set is precisely $(0, \alpha_s/(-\beta_s))$; otherwise it is the whole positive axis. Intersecting these conditions proves the interval assertion and (6).

Suppose the threshold is finite. The source remains strictly positive at ε_*. Its $\mathcal{L}$ is still in $\mathcal{C}_n$, so the ordinary recovered conductances a have finite positive values there. Ordinary recovery and the invertibility in Lemma 6.3 make a and μ continuous at this point. Approaching from the positive interval gives $\mu \geq 0$ in the limit, with at least one zero coordinate; otherwise positivity would continue past the threshold. Equations (32)–(33) now give $r_I \geq 1$, finite strictly positive $a_{uv}r_u r_v$, and finite nonnegative $r_s\mu_s$. Their zero coordinates are exactly the vanishing center edges, as claimed. □

For a concrete computation of (6), let $\alpha_s = f_s(M_0)$. In the edge case, $\beta_s = f_s(M_b(1)) - \alpha_s$. In the spike case,

$$\beta_s = (1+m)f_s(M_b(1)) - \alpha_s.$$

These two evaluations of the original full response determine all coefficients in the threshold formula.

8.1 An exact seven-sided threshold

Consider the $n = 7$ edge family, with source edge p_0p_1 of conductance ε. Specify its common baseline by Table 6; every displayed entry is divided by 100. The separating minors give

$$\varepsilon_* = \frac{54997063108}{61616096925}.$$

The active minor is the one for e_2, with

Inherited edges	Index order	100 times the conductances
$w_{0,i}w_{0,i+1}$	$i = 1, \ldots, 6$	$103, 110, 117, 105, 112, 100$
$w_{1,i}w_{1,i+1}$	$i = 0, \ldots, 6$	$107, 114, 102, 109, 116, 104, 111$
$w_{0,i}w_{1,i}$	$i = 0, \ldots, 6$	$118, 106, 113, 101, 108, 115, 103$
$zw_{1,i}$	$i = 0, \ldots, 6$	$110, 117, 105, 112, 100, 107, 114$

Table 6: The positive baseline for the seven-sided example.

$$P_2 = (1, 2, 3, 4), \qquad Q_2 = (0, 5, 6, z), \qquad \sigma_2 = -1.$$

For this baseline its affine coefficients are

$$\alpha_2 = \frac{791977108969211347}{6438130505517017250000}, \qquad \beta_2 = -\frac{47322321642570149}{343366960294240920000}.$$

All other selected minors have nonnegative slope in this example. At this threshold the center edge e_2 vanishes, while all noncenter conductances stay strictly positive. An exact recovery over $\mathbb{Q}(\varepsilon)$ gives the same value and verifies equality of the full 8×8 response matrices; the computational records are listed in Appendix D.

A The finite algorithm

The input is an integer $n \geq 3$ and a strictly positive source conductance vector c. First decide positive target realizability by (3); on the positive domain, recover the unique target weights and record the divisors used.

Precomputation for n. Construct the three labelled graphs in Table 1 and the ordered pair in Table 2. Fix the boundary-stripping sequence of Proposition 3.5. At each step, enumerate circular connections in the star-free graph in a fixed order and retain the first one broken by that step. Existence of a witness follows from criticality. Existence of each connection is a finite graph problem: enumerate simple paths and test their allowed internal vertices and required disjointness. These graph and witness data are reused for every input on X_n. Also construct the center-edge pairs of Section 7.2 and their signs from (42).

Evaluation at c. Compute the complete response and test the separating minors before performing target recovery.

1. Compute the complete source response $M = \Lambda_{X_n}(c)$ using the internal set specified by the source graph. If any $f_s(M) \leq 0$, return *no strictly positive target*. Otherwise continue with the following recovery steps, recording each divisor. All these divisors are nonzero by Proposition 3.7, applied to the positive target supplied by the criterion.

2. Evaluate the appropriate formula in (16) or (17) at the target position in Table 1. Retain the new conductance η and the stripped response N.

3. Apply the precomputed stripping sequence to N, using (14)–(15) and the exact response updates. Read the final innermost cycle and center conductances from the remaining off-diagonal entries.

4. Assign these values to the inherited target edge labels and assign η to the new feature, obtaining c'.

Return the positive vector c' and the recorded divisors. The inverse rational map is evaluated by computing the target response and using the source completion formulas on their domain of definition, as in Proposition 3.7. For a fixed degeneration ray, the two response evaluations in Section 8, following its main proof, determine α_s, β_s and hence the threshold (6).

B Local response geometry and effective applicability

B.1 The common response hypersurface

Corollary B.1. *The Zariski closure of the response image of H_n is*

$$\mathcal{H}_n = \{M \in \mathcal{V}(B) : F_n(M) = 0\}.$$

This is an irreducible hypersurface of dimension $d - 1$. On a nonempty open subset of it, $\Lambda_{H_n}(\mathcal{R}_n(M)) = M$.

Proof. The row and column sets of F_n are disjoint, so its k^2 entries are distinct independent coordinates in $\mathcal{V}(B)$. The determinant of a generic matrix is irreducible: the singular matrix locus is the closure of the image of products of k-by-$(k-1)$ and $(k-1)$-by-k matrices, and the determinant has multiplicity one at a rank-$(k-1)$ point. For $k = 1$ it is simply a coordinate.

The parameter space of H_n is irreducible. Its rational left inverse in Proposition 3.5 makes the image closure have dimension $|E(H_n)| = d - 1$. Proposition 3.3 places that closure in $\mathcal{H}_n$. Equal dimensions and irreducibility give equality, and the two inverse identities follow on their common rational domain. □

B.2 The common positive response neighborhood

Corollary B.2. *Let $\Omega_Y = \Lambda_Y(\mathbb{R}_{>0}^{E(Y)})$ denote the positive response image. Every $M_0 = \Lambda_{H_n}(b_0)$, $b_0 > 0$, has a neighborhood W in $\mathcal{V}(B)$ such that*

$$\Omega_{X_n} \cap W = \Omega_{X_n'} \cap W = \{M \in W : (-1)^k F_n(M) > 0\}.$$

In particular,

$$\Lambda_{H_n}(\mathbb{R}_{>0}^{E(H_n)}) \subset \partial\big(\Omega_{X_n} \cap \Omega_{X_n'}\big).$$

Proof. The identity $\mathcal{R}_n \circ \Lambda_{H_n} = \mathrm{id}$, with $\mathcal{R}_n$ regular at M_0, shows that $D\Lambda_{H_n}$ has rank $d-1$ at b_0. Its image lies in $\ker dF_n$.

For either completion Y, let $\Phi_Y(s, b)$ be its response with feature conductance s for an edge or s^{-1} for a spike. The boundary formulas extend this map smoothly to $s = 0$, where $\Phi_Y(0, b) = \Lambda_{H_n}(b)$. Lemma 4.1, applied to the feature of Y, gives

$$(-1)^k \frac{\partial}{\partial s} F_n(\Phi_Y(s, b_0))\Big|_{s=0} > 0.$$

Thus $dF_n \neq 0$ at M_0, and the s-direction is transverse to the $d-1$ baseline directions. The inverse function theorem makes Φ_Y a local chart onto a neighborhood W_Y of M_0. Shrink its parameter neighborhood so that $b > 0$. The exact boundary formulas show that $(-1)^k F_n(\Phi_Y(s, b))$ has the sign of s there. Consequently every matrix on the positive side in W_Y is the response of a positive completion. Conversely, every positive completion lies on this side by Proposition 3.3. Take $W = W_{X_n} \cap W_{X_n'}$. The asserted equality follows, and $F_n(M_0) = 0$ places each positive common response on its boundary. □

B.3 Computable applicability conditions

The finite algorithm tests applicability by rational inequalities. We now construct a positive rational input and an open box on which all these inequalities hold.

Lemma B.3. *Given finitely many rational functions defined at a rational point c_*, whose required nonzero values and required signs are known, one can compute a rational open box about c_* preserving them.*

Proof. List the finitely many polynomial numerators and denominators of all required divisions, signs, and determinants; orient each nonzero polynomial so that its value at c_* is positive. Expand

$$p(c_* + u) = p(c_*) + \sum_{\alpha \neq 0} p_\alpha u^\alpha, \qquad A_p = \sum_{\alpha \neq 0} |p_\alpha|.$$

Set $r_p = 1$ if $A_p = 0$, and otherwise

$$r_p = \min\left(1, \frac{p(c_*)}{2A_p}\right).$$

For $\|u\|_\infty < r_p \leq 1$, the nonconstant part has absolute value less than $p(c_*)/2$. Take the minimum of these bounds and, when positive input coordinates are required, $\frac{1}{2}\min_e(c_*)_e$. This is a positive rational radius. □

To obtain c_* effectively, take $b = \mathbf{1}$ and substitute the family $c_{\mathbf{1}}(\varepsilon)$ into the finite formulas. Normalize the feature as η_t/ε for a target edge or $\varepsilon\eta_t$ for a target spike. The normalized feature and all other target conductances have positive limits at 0. For every required polynomial factor remove its power of ε and orient its nonzero constant term positively. The factors controlling recovery divisions have nonzero limits by Proposition 3.5. For each resulting polynomial $p(\varepsilon) = p_0 + \sum_{j\ge1} p_j\varepsilon^j$, $p_0 > 0$, take

$$\delta_p = \begin{cases} 1, & \sum_{j\ge1} |p_j| = 0, \\ \min(1, p_0/(2\sum_{j\ge1} |p_j|)), & \text{otherwise.} \end{cases}$$

Their finite minimum δ_n is positive and rational. Choose $\varepsilon_{\text{test}} = \delta_n/2$. Then $c_* = c_{\mathbf{1}}(\varepsilon_{\text{test}})$ is a rational positive input with positive output and all divisions defined. Lemma B.3 gives the box U_n. In particular, $0 < \varepsilon_{\text{test}} < \varepsilon_*(\mathbf{1})$.

C A three-sided formula and its positive domain

The three-sided move was obtained in [1, Appendix A]. It illustrates the uniform construction in our labels. For $n = 3$, all four source vertices p_0, p_1, p_2, z are observed. Write

$$a = c_{p_0p_1}, \quad b = c_{p_2p_0}, \quad d = c_{p_1p_2}, \qquad u = c_{zp_0}, \quad v = c_{zp_1}, \quad w = c_{zp_2}, \qquad s = b + d + w.$$

The target removes p_0p_1, makes the old p_2 an internal vertex v_2, and attaches the new observed p_2 by a spike of conductance h. The target conductances are

$$\begin{aligned} h &= s + \frac{bd}{a}, & \widehat{b} &= b + \frac{as}{d}, & \widehat{d} &= d + \frac{as}{b}, \\ \widehat{u} &= u - \frac{aw}{d}, & \widehat{v} &= v - \frac{aw}{b}, & \widehat{w} &= w\left(1 + \frac{as}{bd}\right). \end{aligned}$$

The edges carrying $\widehat{b}, \widehat{d}, \widehat{w}$ meet v_2.

Indeed, $F_3(M) = M_{01} = -a$, $\beta_2(M) = M_{02}M_{21} = bd$, and $M_{22} = s$. Equations (17) give

$$h = s + \frac{bd}{a}, \qquad N = M + \frac{a}{bd} M_{:,2}M_{2,:}.$$

The entry N_{01} is zero; the other five off-diagonal entries and the zero row sums give the common graph's Laplacian. Inserting the spike using Lemma 3.1 recovers M.

For strictly positive source conductances, all six target conductances are strictly positive if and only if

$$0 < a < \min\left(\frac{ud}{w}, \frac{vb}{w}\right).$$

Every displayed divisor and the spike pivot are then nonzero. This is the three-sided instance of the threshold in Theorem 2.3.

D Exact computational records

Table 7 summarizes the exact checks accompanying the constructions. Response calculations use rational arithmetic. The programs, their local dependencies, and the recorded certificates are collected in the combined computational archive. Its README gives the directory layout and

Calculation	Recorded range
Missing-connection paths and separators	$n = 3, \dots, 30$
Missing minors and feature formulas	$n = 3, \dots, 14$
Complete transformation, inverse and locality	$n = 3, \dots, 10$
Center-edge staircase paths and deletion cuts	$n = 3, \dots, 40$; 817 cases
Separating minors and linkage calculations	$n = 3, \dots, 16$
Signed center-coordinate matrix	$n = 3, \dots, 10$
Full symbolic ray response and exact threshold	$n = 7$

Table 7: Finite exact checks for the uniform constructions.

complete running commands, and its SHA-256 manifest fixes the supplied files. Both the move and ray modules are included.

The staircase entry program is `verify_wedge_structure.py`; `verify_wedge_witnesses.py` checks the minors and their affine formulas at a third parameter after fitting two values. The program `audit_minor_coordinates.py` constructs the responses for $\mu = e_t$ independently and verifies that the matrix $f_s(M_{e_t})$ is diagonal with positive diagonal entries. Its signs are fixed by the explicit pairing in (42). The seven-sided threshold and baseline coefficients are recorded in `wedge_witness_certificate.json`, and agree with the independent $\mathbb{Q}(\varepsilon)$ recovery certificate.

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